

Math for Engineering

Dr. José L. Pabón, PhD

Class will start soon.

We will pass this course with a
great grade & meet our academic
and professional goals!

- Attendance is required.

Any absence justifications must be addressed with the office of the appropriate staff .

- # Reminder!

We will be courteous, civil to each other.

NO SUCH THING AS AN
OBVIOUS QUESTION

ask ask ask any doubt to clear up.

Reminder

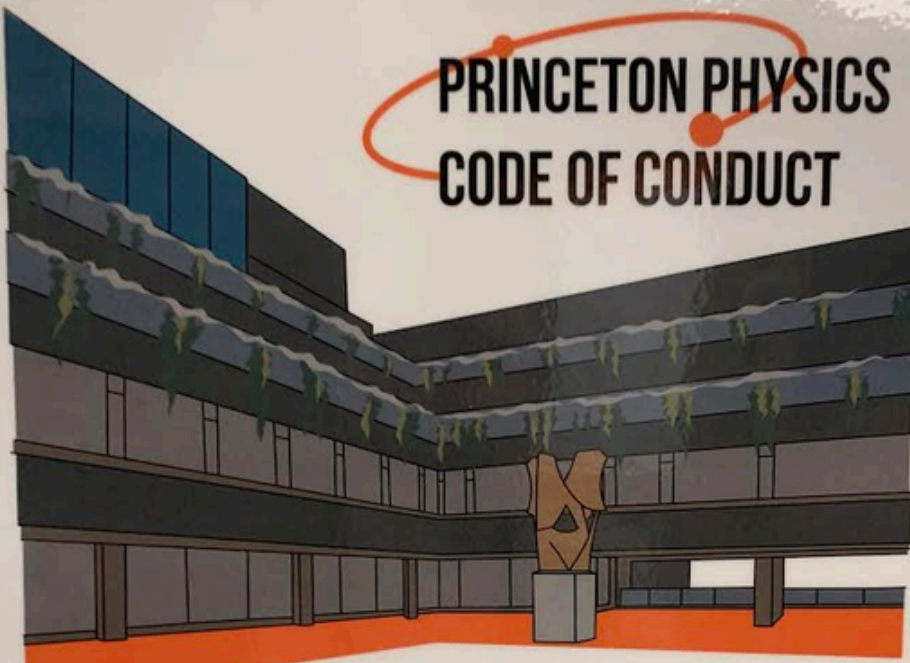
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**PRINCETON PHYSICS
CODE OF CONDUCT**



RESPECT AND SUPPORT COLLEAGUES:

- Be courteous in your interactions with others.
- Refrain from personally critical commentary.
- Give others a chance to voice their thoughts.
- Do not make judgments based on stereotypes.

TAKE INITIATIVE:

- Speak up if others are disrespectful of a group or class of people.

COMMIT TO OPENNESS:

- Be receptive to discussing ways to improve the work environment.
- Do not engage in any overt or perceived retaliation against others.



Integrity – Personal, Academic,
Professional.

P6 section review.

P6 section review.

EXAMPLE 6 Adding and Subtracting Radical Expressions

Simplify: **a.** $\sqrt{45} + 7\sqrt{20}$. **b.** $5\sqrt[3]{80x} - 3\sqrt[3]{270x}$.

Solution

a. We find perfect-square factors for both 45 and 20.

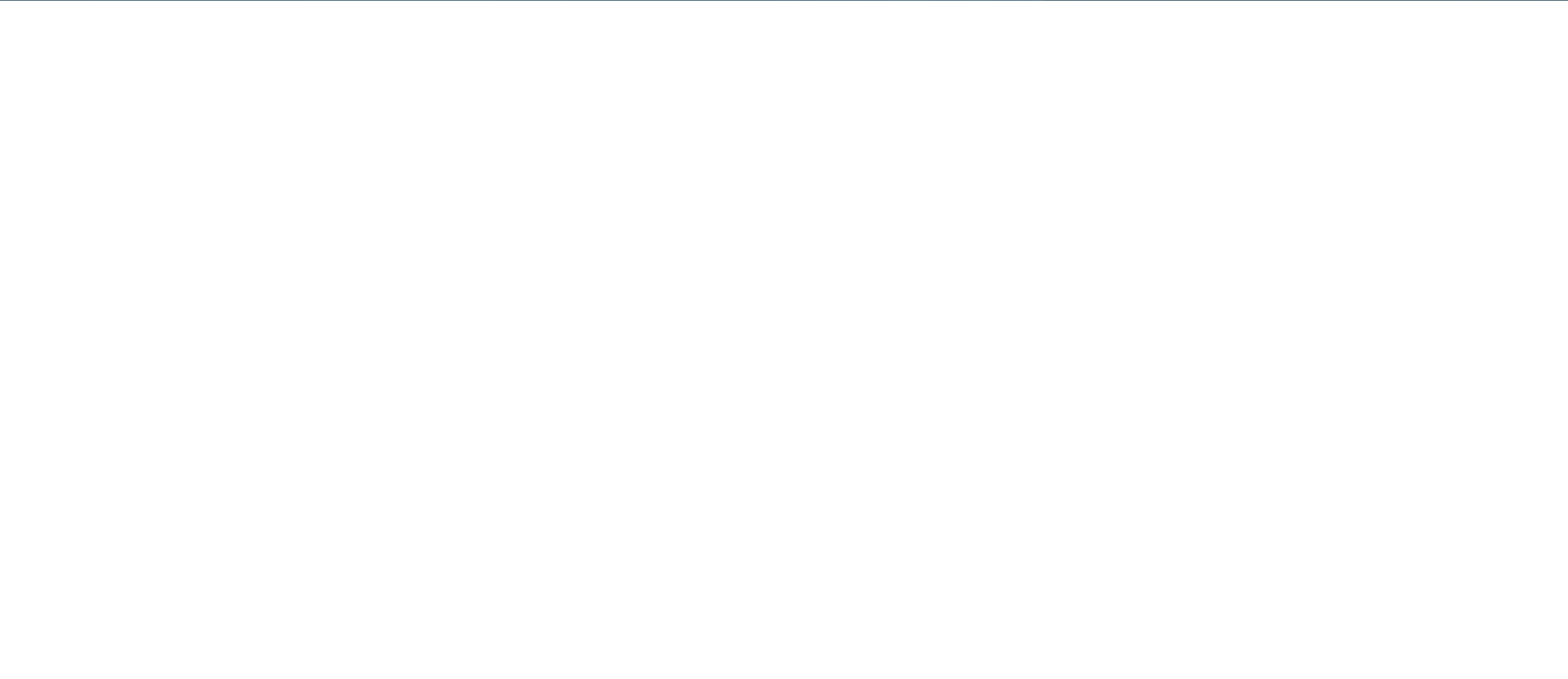
$$\begin{aligned}
 \sqrt{45} + 7\sqrt{20} &= \sqrt{9 \cdot 5} + 7\sqrt{4 \cdot 5} && 45 = 9 \cdot 5; 20 = 4 \cdot 5 \\
 &= \sqrt{9}\sqrt{5} + 7\sqrt{4}\sqrt{5} && \text{Product rule} \\
 &= 3\sqrt{5} + 7 \cdot 2\sqrt{5} && \sqrt{9} = 3; \sqrt{4} = 2 \\
 &= 3\sqrt{5} + 14\sqrt{5} && \text{Like radicals} \\
 &= (3 + 14)\sqrt{5} && \text{Distributive property} \\
 &= 17\sqrt{5}
 \end{aligned}$$

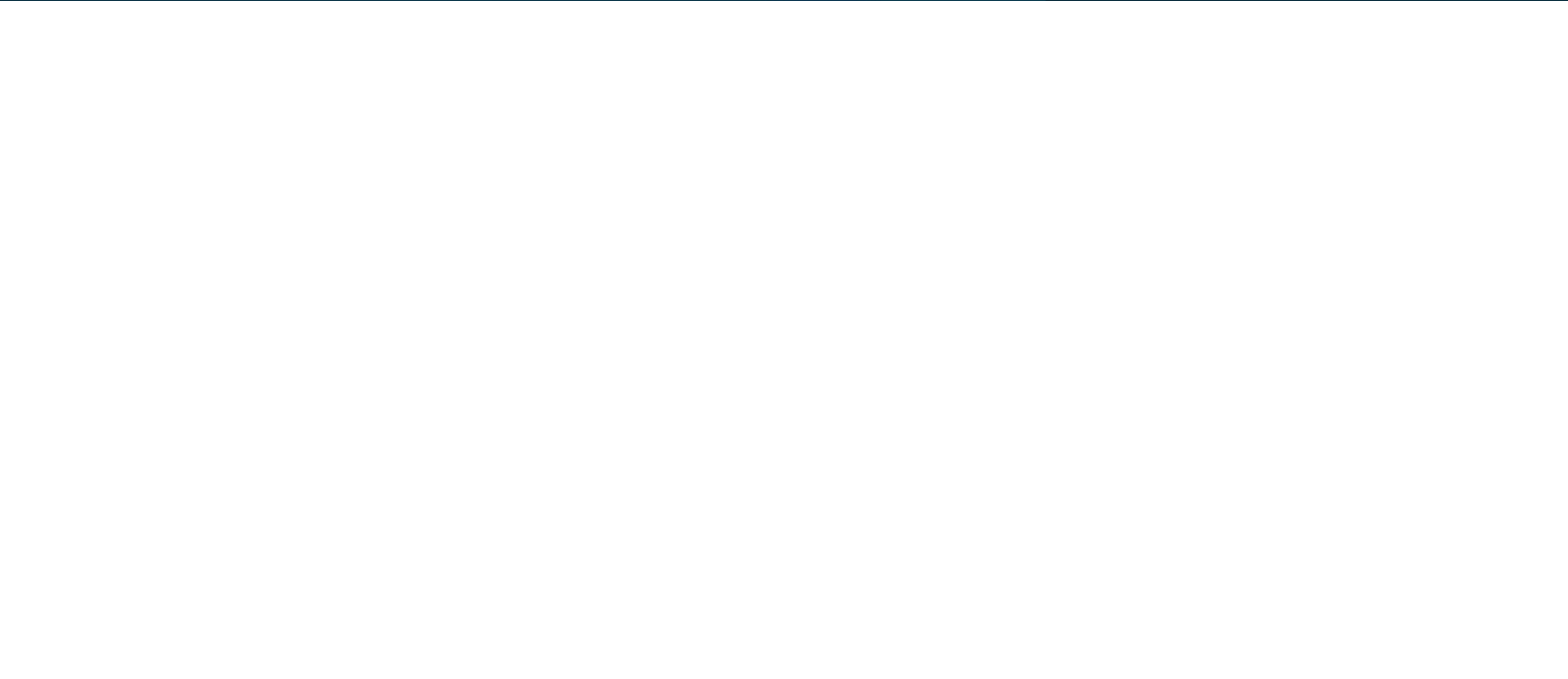
b. This time we find perfect-cube factors for both 80 and 270.

$$\begin{aligned}
 5\sqrt[3]{80x} - 3\sqrt[3]{270x} &= 5\sqrt[3]{8 \cdot 10x} - 3\sqrt[3]{27 \cdot 10x} && 80 = 8 \cdot 10; 270 = 27 \cdot 10 \\
 &= 5\sqrt[3]{8} \cdot \sqrt[3]{10x} - 3\sqrt[3]{27} \cdot \sqrt[3]{10x} && \text{Product rule} \\
 &= 5 \cdot 2 \cdot \sqrt[3]{10x} - 3 \cdot 3\sqrt[3]{10x} && \sqrt[3]{8} = 2; \sqrt[3]{27} = 3 \\
 &= 10\sqrt[3]{10x} - 9\sqrt[3]{10x} && \text{Like radicals} \\
 &= (10 - 9)\sqrt[3]{10x} && \text{Distributive property} \\
 &= \sqrt[3]{10x}
 \end{aligned}$$

Practice Problem 6. Simplify.

a. $3\sqrt{12} + 7\sqrt{3}$ **b.** $2\sqrt[3]{135x} - 3\sqrt[3]{40x}$





Worksheet

Mindfulness

Worksheet

Mindfulness

Discussion and
Solutions





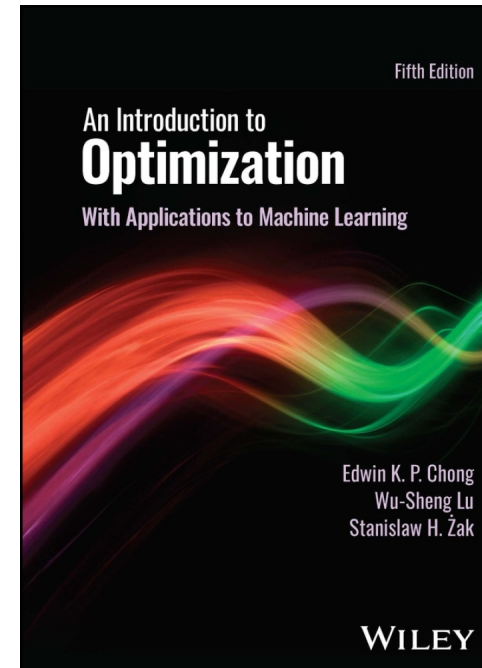
Research Topic - Optimization

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- Mathematical optimization is a branch of math that is about finding a right set of answers that gives the best (optimal) solution to a problem, given some limitations, or criteria.
- In the simplest case, this means that an objective, often called an objective function needs to be minimized or maximized.

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POKÉMON | FINANCE

Pokémon TCG: Chaos Rising Pull Rates

Making sense of Chaos.

By [Peter Day](#) | Published Jun 4, 2026 | 4 min read

The odds on specific cards of each rarity assume an equal population of each card at that rarity. The pull rates for each card we found are within the 95% confidence interval of an individual card of that rarity, which supports the assumption above.

Note: Your personal experience will vary—that's how [randomized booster packs](#) work!

Rarity	Pull Rate (95% Confidence Interval)	Rounded Pull Rate (Per X Packs)
Double Rare	20.30% ± 0.85%	1 in 5 Packs
Ultra Rare	8.29% ± 0.58%	1 in 12 Packs
Illustration Rare	10.66% ± 0.65%	1 in 9 Packs
Special Illustration Rare	1.21% ± 0.23%	1 in 83 Packs
Mega Hyper Rare	0.10% ± 0.07%	1 in 956 Packs



Research Topic - Optimization

MAGIC: THE GATHERING | MODERN | COMPETITIVE

How Many Surveil Lands Should You Be Playing in Modern MTG?

Running the numbers on surveil 1.

By [Frank Karsten](#) | [@karsten_frank](#) | Updated Feb 13, 2025 | 11 min read

Number of surveil lands	Raw metagame share	Non-mirror winrate
0	53.9%	46.1%
1	19.5%	52.0%
2	22.4%	52.2%
3	2.4%	54.4%
4	1.9%	55.4%



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Surveil Lands in Dimir Mill



Market Price: \$2.83



Market Price: \$3.40

Out of all Dimir Mill players:

- 48.4% used 0 surveil lands, with a 66-67 (49.6%) record.
- 33.9% used 1 surveil land, with a 47-62 (43.1%) record.
- 17.7% used 2 surveil lands, with a 31-25 (55.4%) record.

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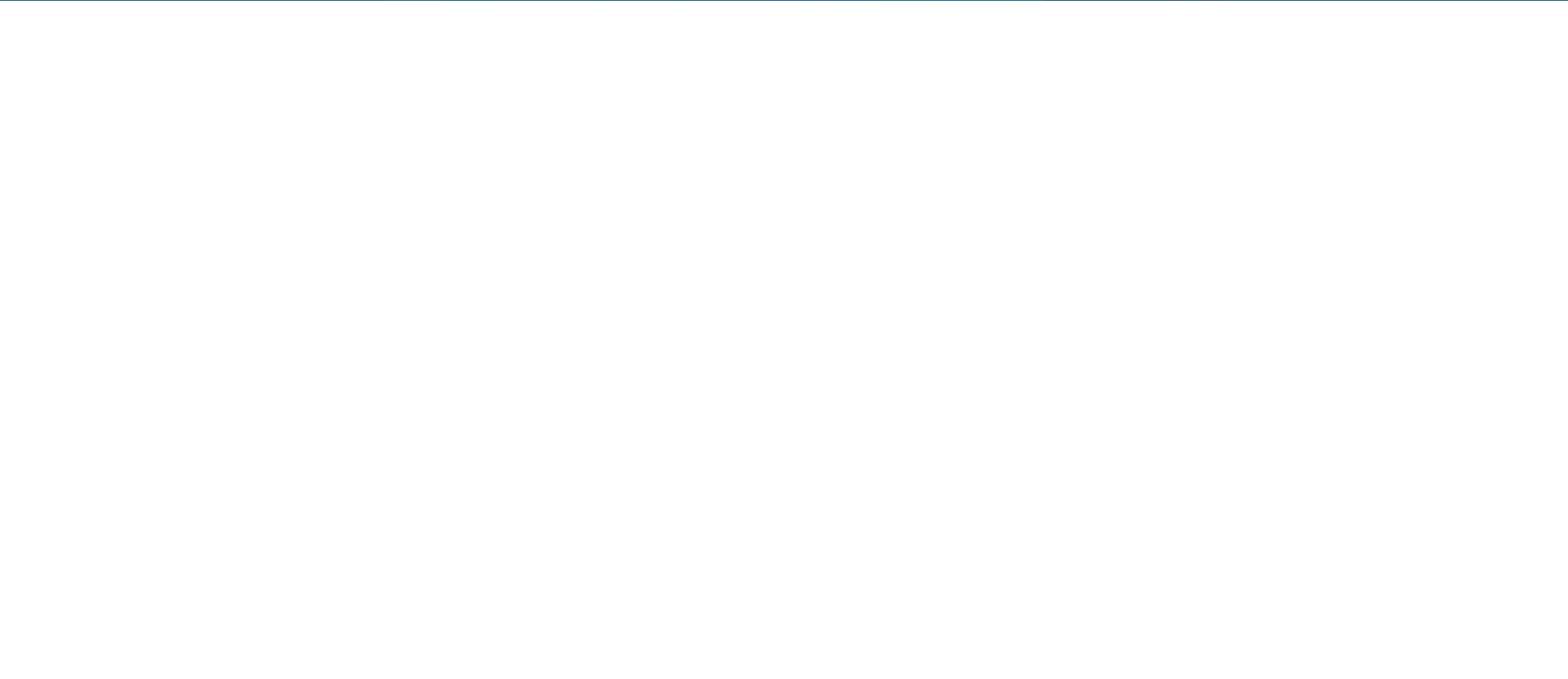


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- The article analyzed the win rates of nearly 6,000 competitive Modern decks from premier competitive tournaments in February and March of year 2025.
- In general, one surveil land is the sweet spot for most fair multicolor Modern decks, and a second surveil land can be worth it for combo strategies.
- Archetypes that rely on graveyards or cascade spells, such as non-DomRhinos, Living End or Goryo's Vengeance, may want to run three or four, but you need good reasons to do so.
- Overall, surveil lands were a major addition to the format, and players who adopted them quickly reaped the rewards.



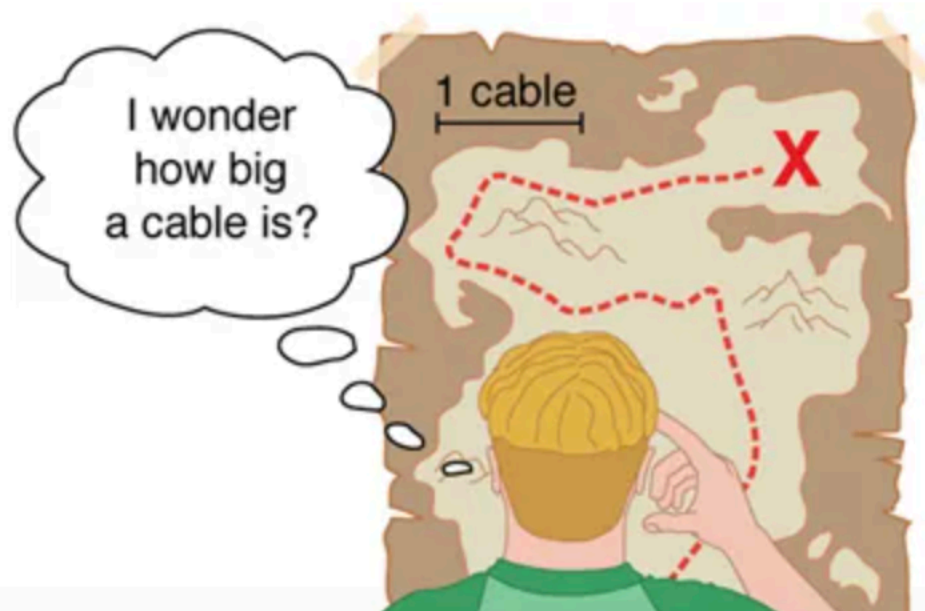
Introduction - reading

Introduction - reading

Worksheet!

Define a **physical quantity** either by *specifying how it is measured* or by *stating how it is related* from other measurements. For example, we define distance and time by giving methods for measuring them, whereas we define *average speed* by stating that it is calculated as distance traveled divided by time of travel.

Measurements of physical quantities are expressed in terms of **units**, which are standardized values. For example, the length of a race, which is a physical quantity, can be expressed in units of meters (for sprinters) or kilometers (for distance runners). Without standardized units, it would be extremely difficult for scientists to express and compare measured values in a meaningful way. (See [Figure 1.16](#).)





There are two major systems of units used in the world: **SI units** (also known as the metric system) and **English units** (also known as the customary or imperial system). **English units** were historically used in nations once ruled by the British Empire and are still widely used in the United States. Virtually every other country in the world now uses SI units as the standard; the metric system is also the standard system agreed upon by scientists and mathematicians. The acronym “SI” is derived from the French *Système International*.

Metric vs. Imperial

The world's two major measurement systems

Weight and Volume



1/4 pound
113.4 grams



1/2 gallon
1.89 liters



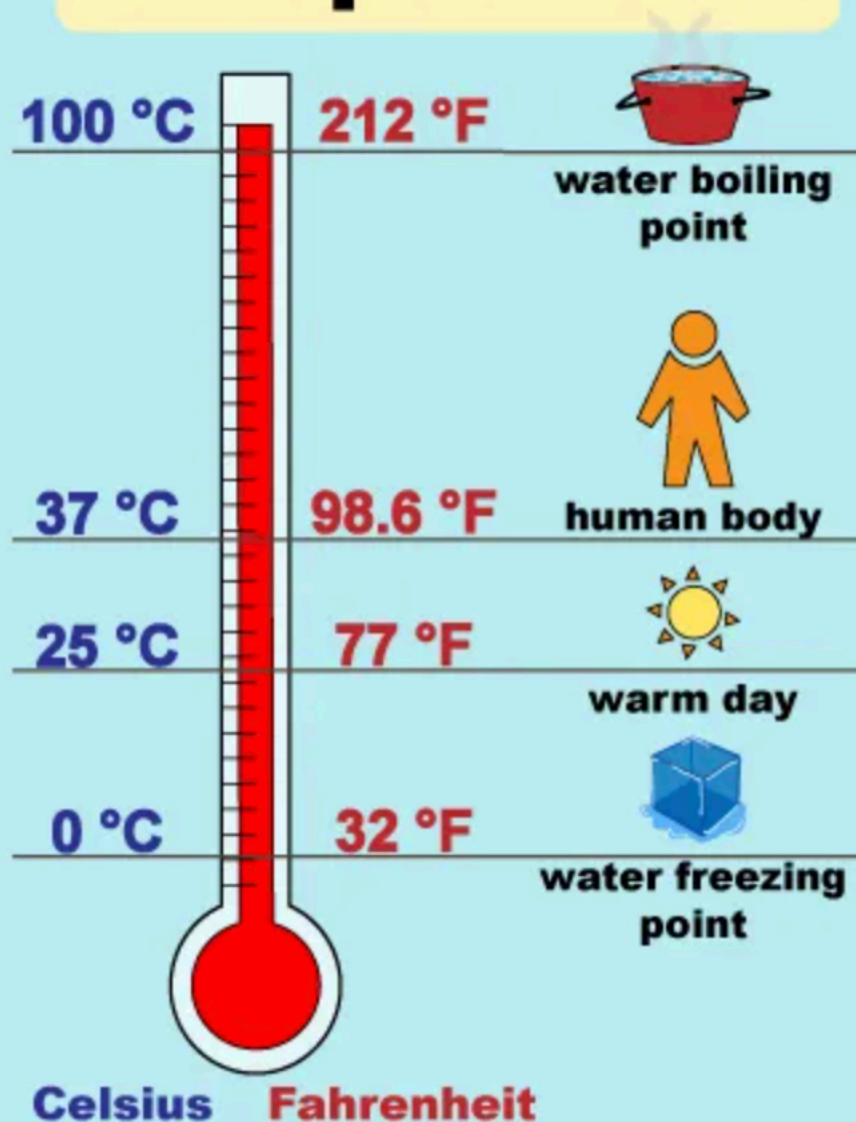
250 grams
0.55 pounds



2 liters
0.53 gallons

The U.K. also uses imperial sizes for milk, including 1, 2, and 6 pints!

Temperature



Car Speeds

**Speed
Limit
60
mph**

U.S. Speed Limit Sign

=

**Speed
Limit
96.56
km/h**

**50
km/h**

Europe Speed Limit Sign

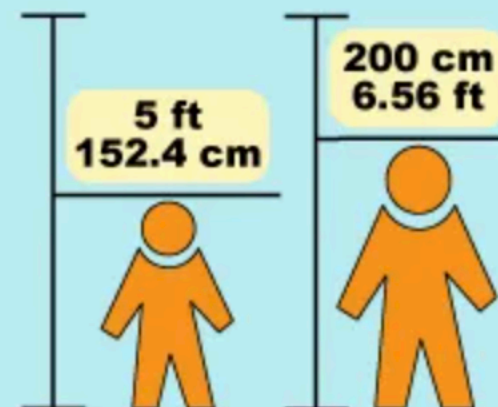
=

**31.07
mph**

Length and Height

1 cm = .39 in.
100 cm = 1 m = 39.4 in.

1 in. = 2.54 cm
12 in. = 1 ft = 30.48 cm



Standard Running Track

400 m $\approx$.25 mile
4 laps = 1600 m $\approx$ 1 mile

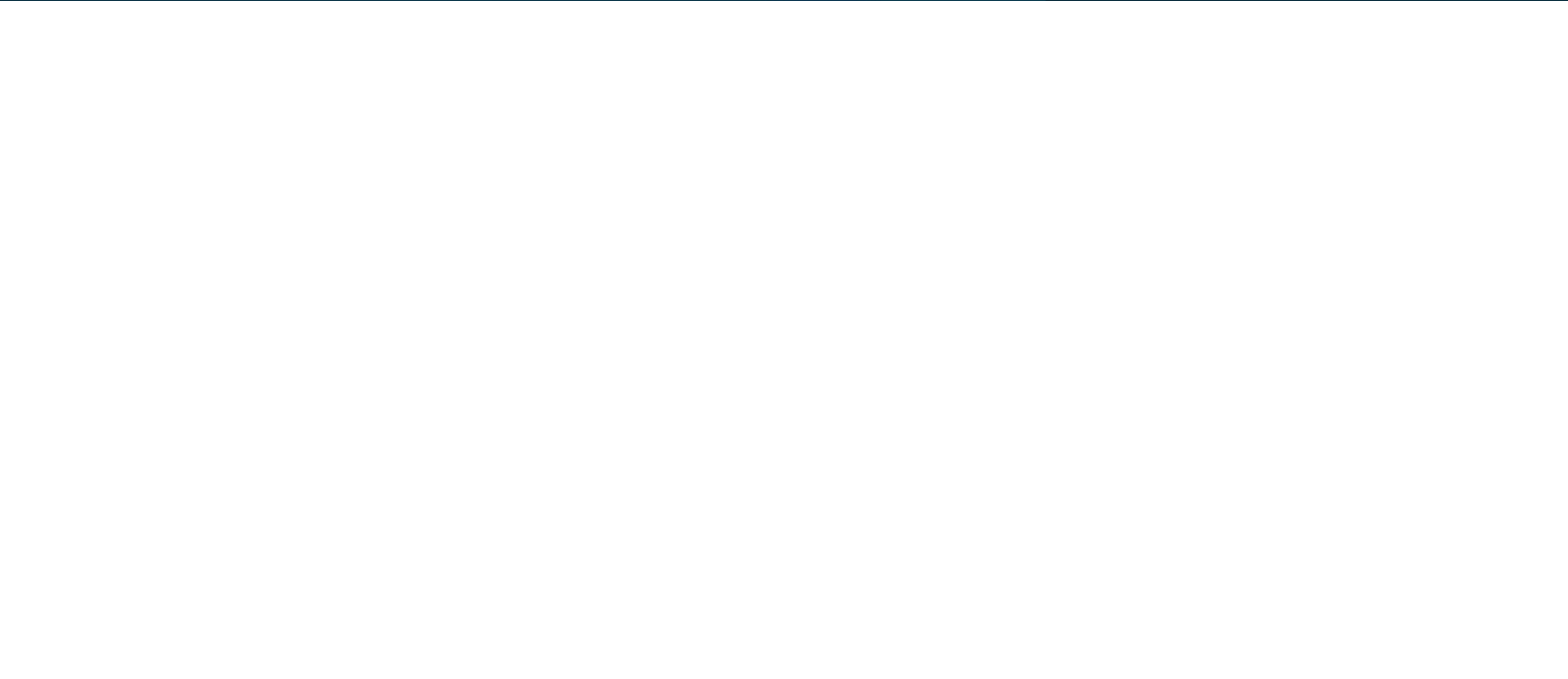
Running tracks are almost universally measured in meters, but the U.S. colloquially treats 1600m as a mile.

Metric vs. Imperial Around the World



Prefix	Symbol	Value	Example (some are approximate)			
exa	E	10^{18}	exameter	Em	10^{18} m	distance light travels in a century
peta	P	10^{15}	petasecond	Ps	10^{15} s	30 million years
tera	T	10^{12}	terawatt	TW	10^{12} W	powerful laser output
giga	G	10^9	gigahertz	GHz	10^9 Hz	a microwave frequency
mega	M	10^6	megacurie	MCi	10^6 Ci	high radioactivity

Length	1 inch (in.) = 2.54 cm (exactly)
	1 foot (ft) = 0.3048 m
	1 mile (mi) = 1.609 km
Force	1 pound (lb) = 4.448 N
Energy	1 British thermal unit (Btu) = 1.055×10^3 J
Power	1 horsepower (hp) = 746 W
Pressure	1 lb/in ² = 6.895×10^3 Pa



Accuracy, Precision, and Uncertainty

The degree of accuracy and precision of a measuring system are related to the **uncertainty** in the measurements. Uncertainty is a quantitative measure of how much your measured values deviate from a standard or expected value. If your measurements are not very accurate or precise, then the uncertainty of your values will be very high. In more general terms, uncertainty can be thought of as a disclaimer for your measured values. For example, if someone asked you to provide the mileage on your car, you might say that it is 45,000 miles, plus or minus 500 miles. The plus or minus amount is the uncertainty in your value. That is, you are indicating that the actual mileage of your car might be as low as 44,500 miles or as high as 45,500 miles, or anywhere in between. All measurements contain some amount of uncertainty. In our example of measuring the length of the paper, we might say that the length of the paper is 11 in., plus or minus 0.2 in. The uncertainty in a measurement, A , is often denoted as δA ("delta A "), so the measurement result would be recorded as $A \pm \delta A$. In our paper example, the length of the paper could be expressed as 11 in. $\pm$ 0.2.

Percent Uncertainty

One method of expressing uncertainty is as a percent of the measured value. If a measurement A is expressed with uncertainty, δA , the **percent uncertainty** (%unc) is defined to be

$$\% \text{ unc} = \frac{\delta A}{A} \times 100\%.$$

A grocery store sells 5-lb bags of apples. You purchase four bags over the course of a month and weigh the apples each time. You obtain the following measurements:

Week 1 weight: 4.8 lb

Week 2 weight: 5.3 lb

Week 3 weight: 4.9 lb

Week 4 weight: 5.4 lb

You have previously determined, using additional measurements, that the uncertainty in the average weight of the 5-lb bag is ± 0.4 lb. What is the average weight and percent uncertainty of the bag's weight, given these 4 measurements?

Strategy

The average of the four measured weights is 5.1 lb. Observe that the measured average agrees with the nominal weight of 5 lb within the uncertainty of 0.4 lb. The uncertainty in this value, δA , is 0.4 lb. We can use the following equation to determine the percent uncertainty of the weight:

$$\% \text{ unc} = \frac{\delta A}{A} \times 100\%.$$

Solution

Plug the known values into the equation:

$$\% \text{ unc} = \frac{0.4 \text{ lb}}{5.1 \text{ lb}} \times 100\% = 8\%.$$

1.9

Discussion

We can conclude that the weight of the apple bag is $5 \text{ lb} \pm 8\%$. Consider how this percent uncertainty would change if the bag of apples were half as heavy, but the uncertainty in the weight remained the same. Hint for future calculations: when calculating percent uncertainty, always remember that you must multiply the fraction by 100%. If you do not do this, you will have a decimal quantity, not a percent value.

In science, very large and very small decimal numbers are conveniently expressed in terms of powers of ten, some of which are listed below:

$$\begin{aligned} 10^3 &= 10 \times 10 \times 10 = 1000 & 10^{-3} &= \frac{1}{10 \times 10 \times 10} = 0.001 \\ 10^2 &= 10 \times 10 = 100 & 10^{-2} &= \frac{1}{10 \times 10} = 0.01 \\ 10^1 &= 10 & 10^{-1} &= \frac{1}{10} = 0.1 \\ 10^0 &= 1 \end{aligned}$$

Using powers of ten, we can write the radius of the earth in the following way, for example:

$$\text{Earth radius} = 6\,380\,000\text{ m} = 6.38 \times 10^6\text{ m}$$

The factor of ten raised to the sixth power is ten multiplied by itself six times, or one million, so the earth's radius is 6.38 million meters. Alternatively, the factor of ten raised to the sixth power indicates that the decimal point in the term 6.38 is to be moved six places *to the right* to obtain the radius as a number without powers of ten.

For numbers less than one, negative powers of ten are used. For instance, the Bohr radius of the hydrogen atom is

$$\text{Bohr radius} = 0.000\,000\,000\,0529\text{ m} = 5.29 \times 10^{-11}\text{ m}$$

The factor of ten raised to the minus eleventh power indicates that the decimal point in the term 5.29 is to be moved eleven places *to*

the left to obtain the radius as a number without powers of ten. Numbers expressed with the aid of powers of ten are said to be in **scientific notation**.

Calculations that involve the multiplication and division of powers of ten are carried out as in the following examples:

$$\begin{aligned} (2.0 \times 10^6)(3.5 \times 10^3) &= (2.0 \times 3.5) \times 10^{6+3} = 7.0 \times 10^9 \\ \frac{9.0 \times 10^7}{2.0 \times 10^4} &= \left(\frac{9.0}{2.0}\right) \times 10^7 \times 10^{-4} \\ &= \left(\frac{9.0}{2.0}\right) \times 10^{7-4} = 4.5 \times 10^3 \end{aligned}$$

The general rules for such calculations are

$$\frac{1}{10^n} = 10^{-n} \quad (\text{A-1})$$

$$10^n \times 10^m = 10^{n+m} \quad (\text{Exponents added}) \quad (\text{A-2})$$

$$\frac{10^n}{10^m} = 10^{n-m} \quad (\text{Exponents subtracted}) \quad (\text{A-3})$$

where n and m are any positive or negative number.

Scientific notation is convenient because of the ease with which it can be used in calculations. Moreover, scientific notation provides a convenient way to express the significant figures in a number, as Appendix B discusses.

The number of *significant figures* in a number is the number of digits whose values are known with certainty. For instance, a person's height is measured to be 1.78 m, with the measurement error being in the third decimal place. All three digits are known with certainty, so that the number contains three significant figures. If a zero is given as the last digit to the right of the decimal point, the zero is presumed to be significant. Thus, the number 1.780 m contains four significant figures. As another example, consider a distance of 1500 m. This number contains only two significant figures, the one and the five. The zeros immediately to the left of the unexpressed decimal point are not counted as significant figures. However, zeros located between significant figures are significant, so a distance of 1502 m contains four significant figures.

Scientific notation is particularly convenient from the point of view of significant figures. Suppose it is known that a certain distance is fifteen hundred meters, to four significant figures. Writing the number as 1500 m presents a problem because it implies that only two significant figures are known. In contrast, the scientific notation of 1.500×10^3 m has the advantage of indicating that the distance is known to four significant figures.

When two or more numbers are used in a calculation, the number of significant figures in the answer is limited by the number of significant figures in the original data. For instance, a rectangular garden with sides of 9.8 m and 17.1 m has an area of (9.8 m)(17.1 m). A calculator gives 167.58 m^2 for this product. However, one of the original lengths is known

only to two significant figures, so the final answer is limited to only two significant figures and should be rounded off to 170 m^2 . In general, *when numbers are multiplied or divided, the number of significant figures in the final answer equals the smallest number of significant figures in any of the original factors.*

The number of significant figures in the answer to an addition or a subtraction is also limited by the original data. Consider the total distance along a biker's trail that consists of three segments with the distances shown as follows:

	2.5 km
	11 km
	<u>5.26 km</u>
Total	18.76 km

The distance of 11 km contains no significant figures to the right of the decimal point. Therefore, neither does the sum of the three distances, and the total distance should not be reported as 18.76 km. Instead, the answer is rounded off to 19 km. In general, *when numbers are added or subtracted, the last significant figure in the answer occurs in the last column (counting from left to right) containing a number that results from a combination of digits that are all significant.* In the answer of 18.76 km, the eight is the sum of $2 + 1 + 5$, each digit being significant. However, the seven is the sum of $5 + 0 + 2$, and the zero is not significant, since it comes from the 11-km distance, which contains no significant figures to the right of the decimal point.

Exercise 1: Standard Decimal Addition

****Problem:**** A chemist mixes three liquid samples together. Their volumes are measured as 14.25 mL, 7.3 mL, and 0.115 mL. What is the total volume of the mixture reported to the correct number of significant figures?

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****Solution:**** 1. ****Align and Add:**** Write out the standard mathematical sum:

$$\begin{array}{r} 14.25 \text{ mL} \\ 7.3 \text{ mL} \\ + 0.115 \text{ mL} \\ \hline 21.665 \text{ mL} \end{array}$$

2. ****Identify the Limiting Column:**** * 14.25 ends in the hundredths place (2 decimal places).

* 7.3 ends in the ****tenths place**** (1 decimal place). * 0.115 ends in the thousandths place (3 decimal places).

3. ****Round the Answer:**** Because 7.3 is only known to the tenths place, the final answer must be rounded to the tenths place. * 21.665 mL rounds to ****21.7 mL****.

Exercise 2: Addition with Ambiguous/Trailing Zeros

****Problem:**** An engineer is calculating the total length of a pipeline consisting of two sections. The first section is roughly measured to be 1300 m (where the trailing zeros are not significant). The second section is precisely measured to be 255 m. What is the total length?

Exercise 2: Addition with Ambiguous/Trailing Zeros

****Problem:**** An engineer is calculating the total length of a pipeline consisting of two sections. The first section is roughly measured to be 1300 m (where the trailing zeros are not significant). The second section is precisely measured to be 255 m. What is the total length?

****Solution:**** 1. ****Align and Add:****

$$\begin{array}{r} 1300 \text{ m} \\ + 255 \text{ m} \\ \hline 1555 \text{ m} \end{array}$$

2. ****Identify the Limiting Column:**** * Following the procedures learned (like the 1500 m example), the number 1300 m has only two significant figures (1 and 3). This means its last significant digit is in the ****hundreds place****.

* 255 m has its last significant digit in the ones place.

3. ****Round the Answer:**** The final sum is limited by the hundreds place. * 1555 m rounded to the nearest hundred is ****1600 m**** (or 1.6×10^3 m in scientific notation).

Exercise 3: Addition Involving a Radical (Square Root)

****Problem:**** Find the total perimeter length of a custom triangular bracket component by adding the calculated side length $\sqrt{16.0 \text{ cm}^2}$ to an adjacent side length of 3.45 cm.

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****Problem:**** Find the total perimeter length of a custom triangular bracket component by adding the calculated side length $\sqrt{16.0 \text{ cm}^2}$ to an adjacent side length of 3.45 cm.

****Solution:**** 1. ****Step 1: Evaluate the Radical:**** * When taking the square root of a value, the rule of thumb is that the result retains the ****same number of significant figures**** as the original value inside the radical.

* 16.0 cm^2 has ****3 significant figures****. * $\sqrt{16.0} = 4.00 \text{ cm}$ (written with 3 significant figures to preserve accuracy to the hundredths place).

2. ****Step 2: Perform the Addition:****

$$\begin{array}{r} 4.00 \text{ cm} \\ + 3.45 \text{ cm} \\ \hline 7.45 \text{ cm} \end{array}$$

3. ****Identify the Limiting Column:**** * Both 4.00 cm and 3.45 cm extend precisely to the hundredths place (2 decimal places). * Therefore, no additional rounding is needed. The final answer is ****7.45 cm****.

Exercise 1: Density Calculation (Division)

Problem:

An object has a measured mass of 45.20 g and a volume of 15.6 mL. Calculate its density using the formula:

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

Report the answer with the correct number of significant figures based on the guidelines.

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Solution:

1. Identify Significant Figures:

- Mass = 45.20 g has 4 significant figures (the trailing zero after the decimal point is significant).
- Volume = 15.6 mL has 3 significant figures.

2. Perform the Calculation:

$$\text{Density} = \frac{45.20 \text{ g}}{15.6 \text{ mL}} \approx 2.89743589 \text{ g/mL}$$

3. **Apply the Rule:** According to the rule for multiplication and division, the final answer is limited by the factor with the smallest number of significant figures. Here, the volume (15.6 mL) has the fewest (3 significant figures).

4. **Round the Final Answer:** Rounding 2.89743589... to 3 significant figures yields:

$$2.90 \text{ g/mL}$$

A chemical solution is prepared, and its final concentration needs to be determined using the following formula:

$$\text{Concentration} = \frac{\text{Mass of solute} \times \text{Purity factor}}{\text{Volume of solution}}$$

Given that the mass of the solute is 3.50 g, the purity factor is 0.98, and the volume of the solution is 0.2500 L, calculate the concentration to the correct number of significant figures.

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Solution:

1. Identify Significant Figures:

- Mass of solute = 3.50 g has 3 significant figures.
- Purity factor = 0.98 has 2 significant figures (leading zeros are not significant).
- Volume = 0.2500 L has 4 significant figures (trailing zeros after a decimal point are significant).

2. Perform the Calculation:

$$\text{Concentration} = \frac{3.50 \times 0.98}{0.2500} = \frac{3.43}{0.2500} = 13.72 \text{ g/L}$$

3. Apply the Rule: The rule for multiplication and division states that the final result cannot have more significant figures than the least precise factor. The purity factor (0.98) has the fewest significant figures (2).

4. Round the Final Answer: Rounding 13.72 to 2 significant figures yields:

Problem:

The final velocity v of an object dropping from rest over a distance d is determined by the formula:

$$v = \sqrt{2gd}$$

Calculate the final velocity if the acceleration due to gravity is $g = 9.81 \text{ m/s}^2$ and the distance is $d = 12.50 \text{ m}$. (Note: The number 2 is an exact mathematical constant and does not limit the significant figures).

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Solution:**1. Identify Significant Figures:**

- Acceleration (g) = 9.81 m/s^2 has 3 significant figures.
- Distance (d) = 12.50 m has 4 significant figures.
- The constant 2 is exact and has an infinite number of significant figures.

2. Perform the Calculation: First, compute the product inside the radical:

$$2 \times 9.81 \times 12.50 = 245.25$$

Next, extract the square root:

$$v = \sqrt{245.25} \approx 15.6604598 \text{ m/s}$$

3. Apply the Rule: For operations containing roots or radicals, the result inherits the precision limitations of the values inside. Since the core multiplication inside is constrained by g (3 significant figures), the value under the root effectively carries 3 significant figures of precision, meaning the final answer must be rounded to 3 significant figures.

4. Round the Final Answer: Rounding $15.6604598 \dots$ to 3 significant figures yields:

$$15.7 \text{ m/s}$$

Physics systems, solutions

Physics systems, solutions

In order to analyze and determine solutions of Physics systems, we need to understand:

- Linear equations, also known as linear functions.
- The slope of the lines, the standard form.
- The negative reciprocal slopes.
- Parallel lines.
- Perpendicular lines.

Linear Functions A function of the form $f(x) = mx + b$, where m and b are fixed constants, is called a **linear function**. Figure 1.14a shows an array of lines $f(x) = mx$. Each of these has $b = 0$, so these lines pass through the origin. The function $f(x) = x$ where $m = 1$ and $b = 0$ is called the **identity function**. Constant functions result when the slope is $m = 0$ (Figure 1.14b).

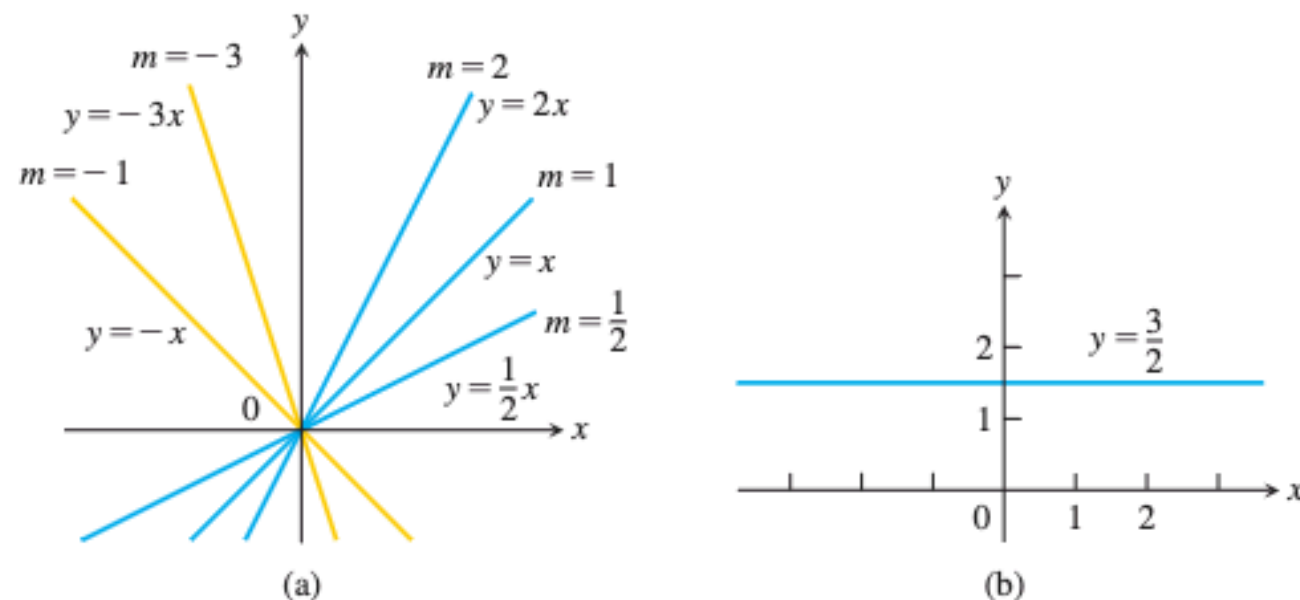


FIGURE 1.14 (a) Lines through the origin with slope m . (b) A constant function with slope $m = 0$.

DEFINITION Two variables y and x are **proportional** (to one another) if one is always a constant multiple of the other—that is, if $y = kx$ for some nonzero constant k .

P3 section review.

Special-Product Formulas

A and B represent any algebraic expressions.

Formula	Example
<i>Product giving a difference of squares</i>	
$(A + B)(A - B) = A^2 - B^2$	$(5x + 2)(5x - 2) = (5x)^2 - 2^2$ $= 25x^2 - 4$
<i>Squaring a binomial sum or difference</i>	
$(A + B)^2 = A^2 + 2AB + B^2$	$(3x + 2)^2 = (3x)^2 + 2(3x)(2) + 2^2$ $= 9x^2 + 12x + 4$
$(A - B)^2 = A^2 - 2AB + B^2$	$(2x - 5)^2 = (2x)^2 - 2(2x)(5) + 5^2$ $= 4x^2 - 20x + 25$
<i>Cubing a binomial sum or difference</i>	
$(A + B)^3 = A^3 + 3A^2B + 3AB^2 + B^3$	$(x + 5)^3 = x^3 + 3x^2(5) + 3x(5)^2 + 5^3$ $= x^3 + 15x^2 + 75x + 125$
$(A - B)^3 = A^3 - 3A^2B + 3AB^2 - B^3$	$(x - 4)^3 = x^3 - 3x^2(4) + 3x(4)^2 - 4^3$ $= x^3 - 12x^2 + 48x - 64$
<i>Products giving a sum or difference of cubes</i>	
$(A + B)(A^2 - AB + B^2) = A^3 + B^3$	$(x + 2)(x^2 - 2x + 4) = x^3 + 2^3 = x^3 + 8$
$(A - B)(A^2 + AB + B^2) = A^3 - B^3$	$(x - 3)(x^2 + 3x + 9) = x^3 - 3^3 = x^3 - 27$

P3 section review.

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$(A + B)^3 = A^3 + 3A^2B + 3AB^2 + B^3$	$(x + 5)^3 = x^3 + 3x^2(5) + 3x(5)^2 + 5^3$ $= x^3 + 15x^2 + 75x + 125$
$(A - B)^3 = A^3 - 3A^2B + 3AB^2 - B^3$	$(x - 4)^3 = x^3 - 3x^2(4) + 3x(4)^2 - 4^3$ $= x^3 - 12x^2 + 48x - 64$
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1.1 section review.

Solving Linear Equations in One Variable

Linear Equations

A **conditional linear equation in one variable**, such as x , is an equation that can be written in the **standard form**

$$ax + b = 0,$$

where a and b are real numbers with $a \neq 0$.

SIDE NOTE

When finding the domain of a variable, remember that

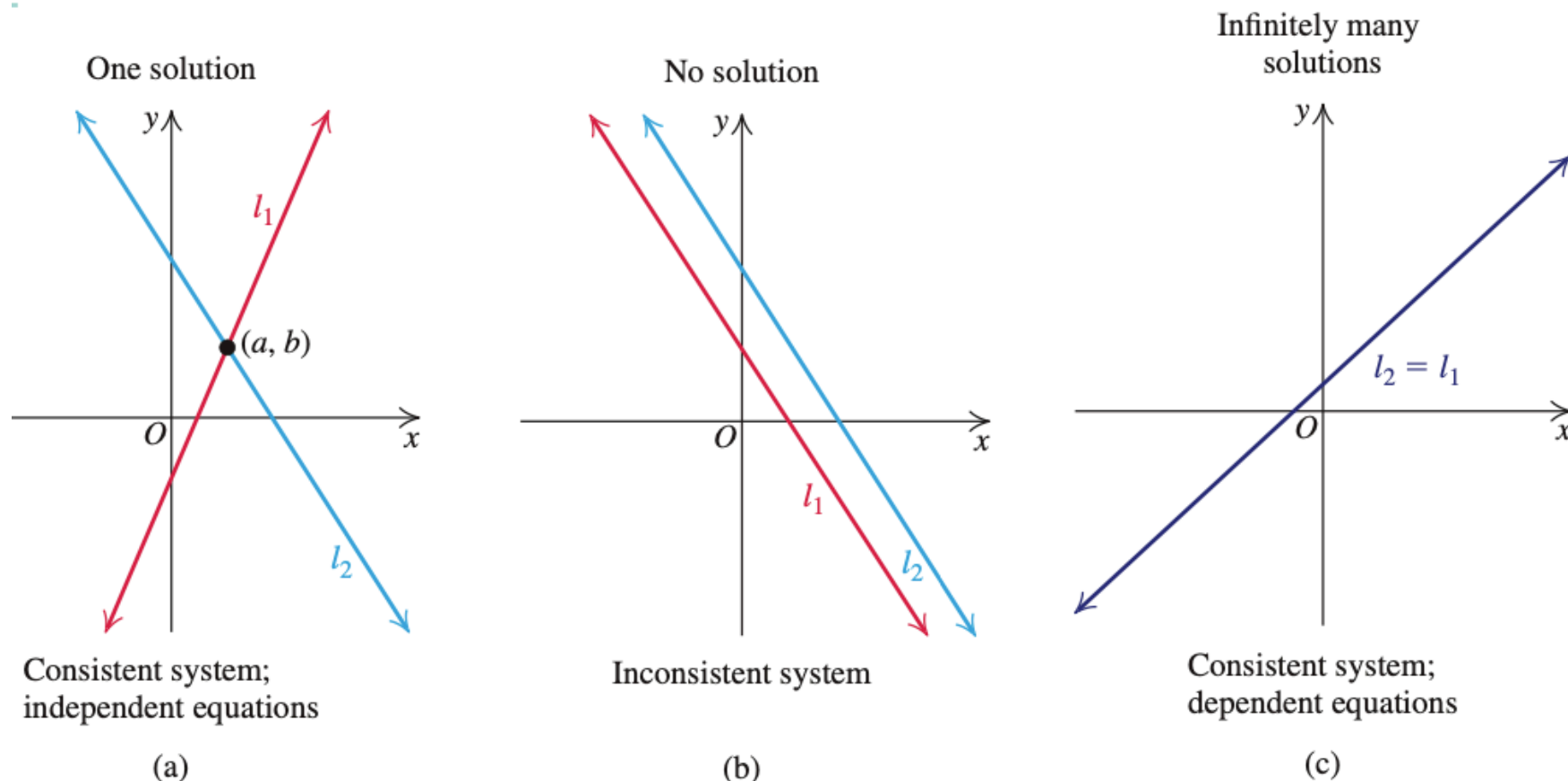
1. Division by 0 is undefined.
2. The square root (or any even root) of a negative number is not a real number.

Identifying Types of Linear Equations

1. An equation that is satisfied by all values in the domain of the variable is an **identity**. For example, $2(x - 1) = 2x - 2$. When you try to solve an identity, you get a true statement that does not depend on the variable, such as $3 = 3$ or $0 = 0$.
2. An equation that is not an identity but is satisfied by at least one number in the domain of the variable is a **conditional** equation. For example, $2x = 6$ is a linear conditional equation; $x = 3$ is a solution, but $x = 0$ is not.
3. An equation that is not satisfied by any value of the variable is an **inconsistent** equation. For example, $x = x + 2$ is an inconsistent equation. Because no number is 2 more than itself, the solution set of the equation $x = x + 2$ is $\emptyset$. When you try to solve an inconsistent equation, you obtain a false statement such as $0 = 2$.

1.1 section

1. One solution, also called **unique solution** (the lines intersect; see Figure 8.2(a)): the system is consistent, and the equations in the system are said to be **independent**.
2. No solution (the lines are parallel; see Figure 8.2(b)): The system is inconsistent.
3. Infinitely many solutions (the lines coincide; see Figure 8.2(c)): The system is consistent, and the equations in the system are said to be **dependent**.



(a) $l_1 \rightarrow y = x - 1, l_2 \rightarrow y = -x + 2,$

(b) $l_1 \rightarrow y = -x, l_2 \rightarrow y = -x + 2,$

(c) $l_1 = l_2 := y = x.$

1.1 section

PROCEDURE IN ACTION

EXAMPLE 3 The Substitution Method

Objective

Reduce the solution of the system to the solution of one equation in one variable by substitution.

Step 1 ▶ Solve for one variable. Choose one of the equations and express one of its variables in terms of the other variable.

Step 2 ▶ Substitute. Substitute the expression found in Step 1 into the other equation to obtain an equation in one variable.

Step 3 ▶ Solve the equation obtained in Step 2.

Step 4 ▶ Back-substitute. Substitute the value(s) you found in Step 3 back into the expression you found in Step 1. The result is the solution(s).

Step 5 ▶ Check. Check your answer(s) in the original equations.

Example

Solve the system:

$$\begin{cases} 2x - 5y = 3 & (1) \\ y - 2x = 9 & (2) \end{cases}$$

1. We choose equation (2) and express y in terms of x .

$$y = 2x + 9 \quad (3)$$

Solve equation (2) for y .

2. $2x - 5y = 3$

Equation (1)

$$2x - 5(2x + 9) = 3$$

Substitute $2x + 9$ for y .

$$2x - 10x - 45 = 3$$

Distributive property

3. $-8x - 45 = 3$

Combine like terms.

$$-8x = 3 + 45$$

Add 45 to both sides.

$$x = -6$$

Solve for x .

4. $y = 2x + 9$

Equation (3) from Step 1

$$y = 2(-6) + 9 = -3$$

Substitute $x = -6$.

Because $x = -6$ and $y = -3$, the solution set is $\{(-6, -3)\}$.

5. Check: $x = -6$ and $y = -3$.

Equation (1)

Equation (2)

$$2(-6) - 5(-3) \stackrel{?}{=} 3$$

$$-3 - 2(-6) \stackrel{?}{=} 9$$

$$-12 + 15 = 3 \checkmark$$

$$-3 + 12 = 9 \checkmark$$

Practice Problem 3. Solve: $\begin{cases} x - y = 5 & (1) \\ 2x + y = 7 & (2) \end{cases}$

1.1 section review.

104. Investment. Mr. Sharma invested a total of \$30,000 in two ventures for a year. The annual return from one of them was 8%, and the other paid 10.5% for the year. He received a total income of \$2550 from both investments. How much did he invest at each rate?

1.1 section.

104. Investment. Mr. Sharma invested a total of \$30,000 in two ventures for a year. The annual return from one of them was 8%, and the other paid 10.5% for the year. He received a total income of \$2550 from both investments. How much did he invest at each rate?

- x (investment 1) + y (investment 2) = ?

1.1 section.

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- x (investment 1 in \$) + y (investment 2 in \$) = \$30000.

8.1 section.

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- x (investment 1 in \$) + y (investment 2 in \$) = \$30000.
- $.08x + .105y = \$2550$.
- Then: $8x + 10.5y = \$255000$.

8.1 section.

104. Investment. Mr. Sharma invested a total of \$30,000 in two ventures for a year. The annual return from one of them was 8%, and the other paid 10.5% for the year. He received a total income of \$2550 from both investments. How much did he invest at each rate?

- x (investment 1 in \$) + y (investment 2 in \$) = \$30000.
- $.08x + .105y = \$2550$.
- Then: $8x + 10.5y = \$255000$.
- Substitute: $8(\$30000 - y) + 10.5y = \255000 .
- $\$240000 - 8y + 10.5y = \255000 .
- $2.5y = \$15000$.
- $y = \$6000$.
- Substitute into first, total equation:
- $x + \$6000 = \30000 .
- $x = \$24000$.

Motivation.

What is mathematics?

What is math?

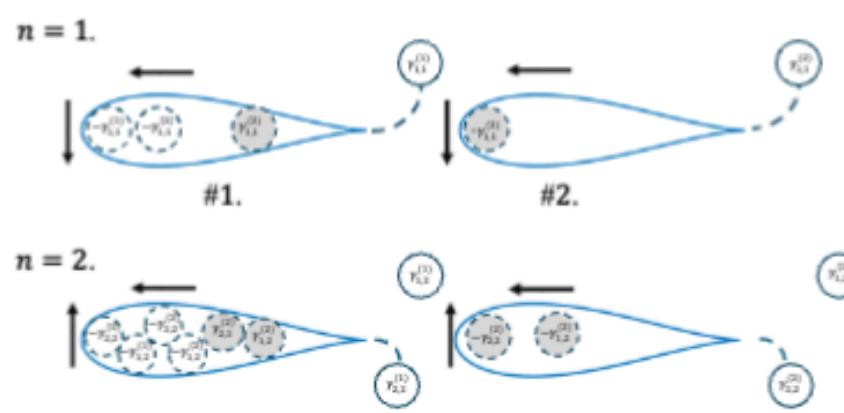


Figure 2.3 Schematic of the flow-induced model configuration for $M = 2$ wings and their shed vortices in two time steps ($n = 1$ and $n = 2$). The velocity potential around wing 1 is considered, so the second wing is only represented by its bound vorticity (gray circles). The dotted circles represent image vortices. The dashed curves connect the trailing edges of each wing to the vortex most recently shed. The bound vorticity before the vortex is shed is $\Gamma_{1,1}^{(1)} = 0$, $\Gamma_{1,1}^{(2)} = 0$.

step is

$$\begin{aligned}
 w_n^{(j)}(\zeta) &= \overline{W}_n^{(j)}(J(\zeta) - \zeta) - \frac{r^2 W_n^{(j)}}{\zeta} - i \frac{1}{2\pi} \sum_{l=1}^M \sum_{k=1}^n \gamma_{k,n}^{(l)} \log \left(\frac{-r}{\zeta_{k,n}^{(j,l)}} \frac{\zeta - \zeta_{k,n}^{(j,l)}}{\zeta - r^2 / \bar{\zeta}_{k,n}^{(j,l)}} \right) \\
 &\quad - i \frac{1}{2\pi} \sum_{l \neq j}^M \left(\Gamma_n^{(l)} - \gamma_n^{(l)} \right) \log \left(\frac{-r}{\chi_n^{(j,l)}} \frac{\zeta - \chi_n^{(j,l)}}{\zeta - r^2 / \bar{\chi}_n^{(j,l)}} \right) \\
 &\quad - i \frac{1}{2\pi} \left[\sum_{l=1}^M \left(\Gamma_n^{(l)} + \sum_{k=1}^{n-1} \gamma_{k,n}^{(l)} \right) \right] \log \left(\frac{\zeta}{r} \right), \\
 \zeta_{k,n}^{(j,l)} &= J^{-1} \left(z_{k,n}^{(l)} - Z_n^{(j)} \right), \quad \chi_n^{(j,l)} = J^{-1} \left(Z_n^{(l)} - Z_n^{(j)} \right).
 \end{aligned} \tag{2.23}$$

That is, $\zeta_{k,n}^{(j,l)}$ is the position of vortex $z_{k,n}^{(l)}$ at time n in the ζ -plane, which was shed by wing l at time step k , using coordinates where wing j is at the center. Similarly, $\chi_n^{(j,l)}$ is the position of wing l at time n , again using coordinates where wing j is at the

Motivation.

What is mathematics?

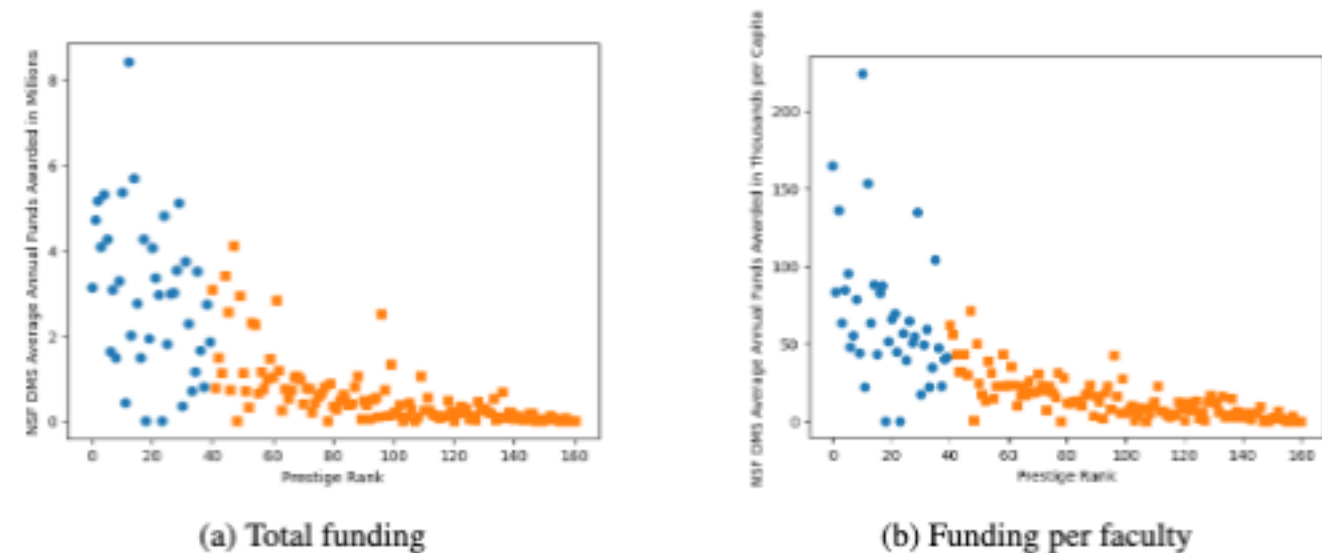


Fig. 4: Average annual grant funding for Mathematics departments by the prestige rank of the department, displayed by total funding to the department (a), and on a per capita basis accounting for the varying number of faculty in each department (b).

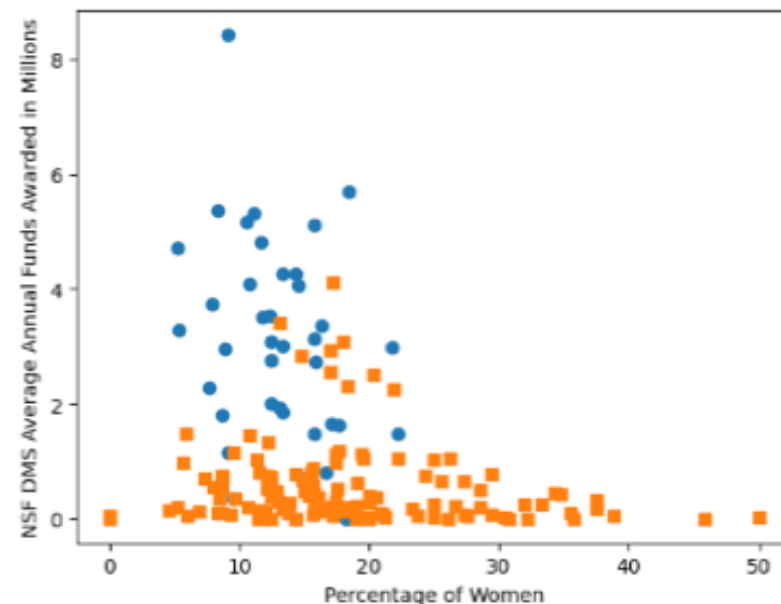


Fig. 5: Average annual grant funding for Mathematics departments by the percentage of women faculty in the department. The upper quartile (by prestige) departments are represented by blue circles, while the remainder are orange squares.

Motivation.

What does math have to do with engineering?

Can we think of different kinds of engineers?

How does math scaffold in the daily duties of an engineer ?

Presentation and communication.

Determine the values of 'x' that satisfy this equation:

$$216x^3 - 8 = 0,$$

Presentation and communication.

Determine and describe the reasons the Roman Empire dominated their regions from 27 BC to 476 AC?

Presentation and communication.

Determine and describe the reasons the Roman Empire dominated their regions from 27 BC to 476 AC?

The Roman Empire dominated its regions through a combination of overwhelming military superiority, highly organized administrative infrastructure, and strategic cultural assimilation. Professionally trained legions and advanced engineering allowed Rome to conquer and secure vast territories, while an extensive network of roads and centralized governance facilitated efficient trade and communication. Furthermore, by extending citizenship and incorporating local elites into their administrative framework, the Romans successfully fostered regional stability and long-term loyalty across diverse populations.

Presentation and communication.

Determine the values of 'x' that satisfy this equation:

$$216x^3 - 8 = 0,$$

Presentation and communication.

Determine the values of 'x' that satisfy this equation: $216x^3 - 8 = 0,$

$$8 (27x^3 - 1) = 0,$$

$$(27x^3 - 1) = 0,$$

$$(3x - 1) (3x^2 + 6x + 1) = 0,$$

$$\implies (3x - 1) = 0, (3x^2 + 6x + 1) = 0,$$

$$\implies x = \left\{ \frac{1}{3}, \frac{\sqrt{6}}{3} - 1, -\frac{\sqrt{6}}{3} - 1 \right\}$$

Presentation and communication.

Determine the values of 'x' that satisfy this equation:

$$216x^3 - 8 = 0,$$

$$8 (27x^3 - 1) = 0, \text{ Distributive Prop.}$$

$$(27x^3 - 1) = 0, \text{ Mult. eq by } \frac{1}{8}$$

$$(3x - 1) (3x^2 + 6x + 1) = 0, \text{ Polynom. Factor.}$$

$$\implies (3x - 1) = 0, (3x^2 + 6x + 1) = 0, \text{ Zero Prop.}$$

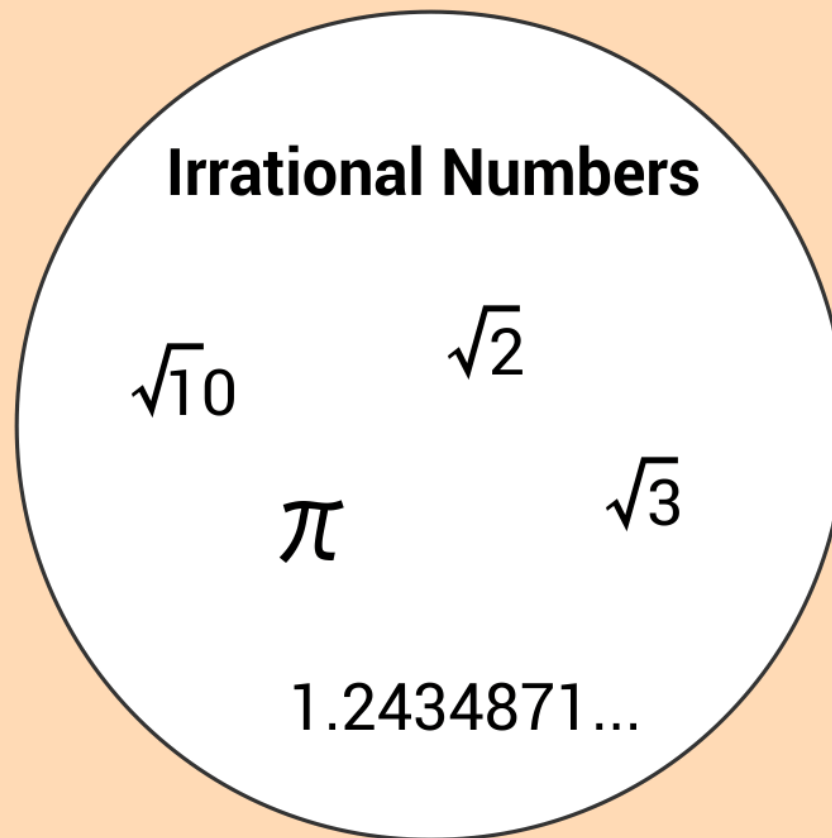
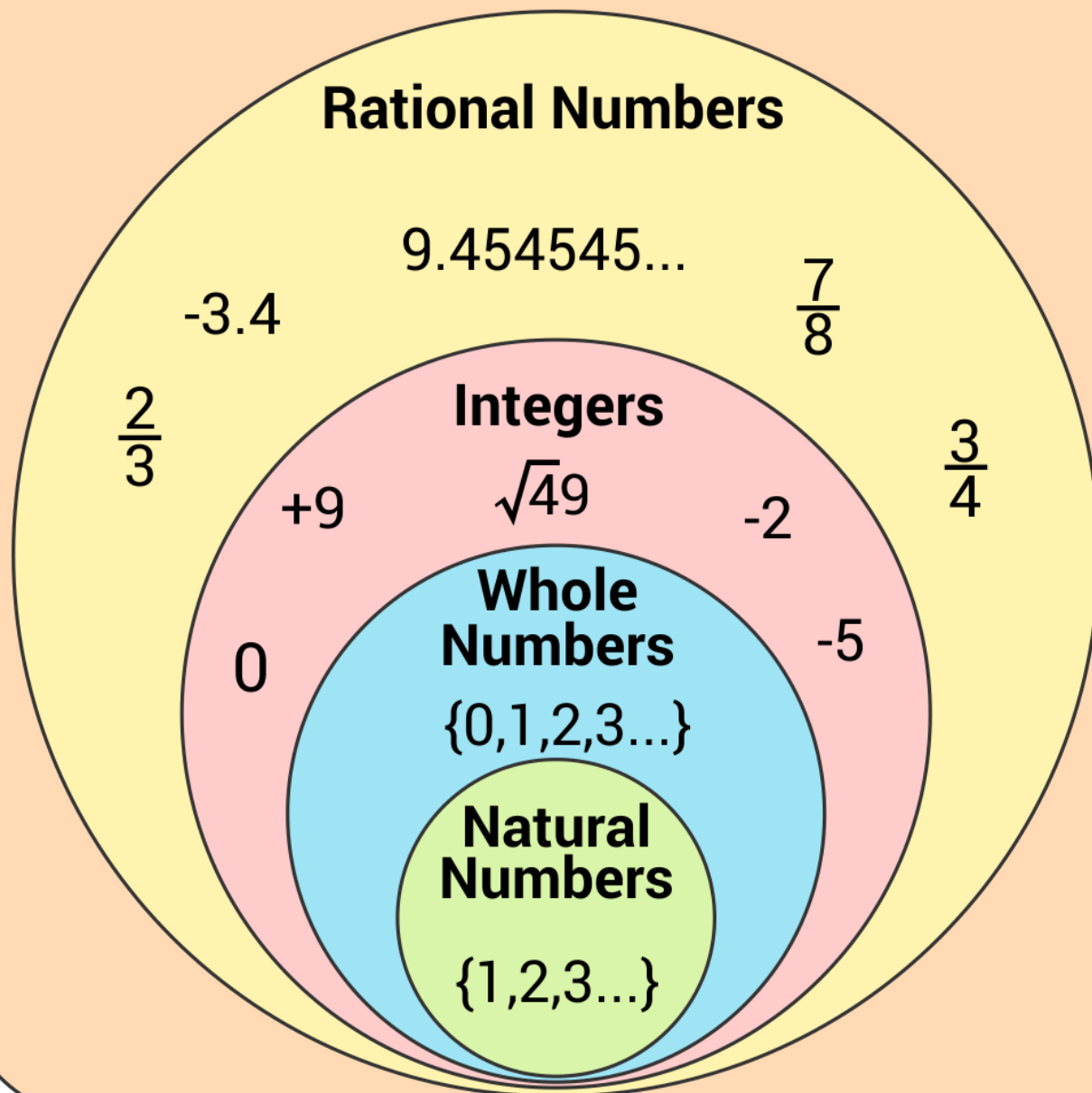
$$\implies x = \left\{ \frac{1}{3}, \frac{\sqrt{6}}{3} - 1, -\frac{\sqrt{6}}{3} - 1 \right\}$$

Review.

Review.

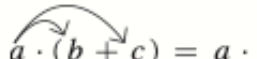
R

Real Numbers



Section 1.1

Properties of the Real Numbers

Name	Addition (A)	Multiplication (M)	Examples
Closure	$a + b$ is a real number	$a \cdot b$ is a real number	A: $1 + \sqrt{2}$ is a real number. M: $2 \cdot \pi$ is a real number.
Commutative	$a + b = b + a$	$a \cdot b = b \cdot a$	A: $4 + 7 = 7 + 4$ M: $3 \cdot 8 = 8 \cdot 3$
Associative	$(a + b) + c = a + (b + c)$	$(a \cdot b) \cdot c = a \cdot (b \cdot c)$	A: $(2 + 1) + 7 = 2 + (1 + 7)$ M: $(5 \cdot 9) \cdot 13 = 5 \cdot (9 \cdot 13)$
Identity	There is a unique real number 0, called the additive identity , such that $a + 0 = a$ and $0 + a = a$.	There is a unique real number 1, called the multiplicative identity , such that $a \cdot 1 = a$ and $1 \cdot a = a$.	A: $3 + 0 = 3$ and $0 + 3 = 3$ M: $7 \cdot 1 = 7$ and $1 \cdot 7 = 7$
Inverse	There is a unique real number $-a$, called the opposite of a , such that $a + (-a) = 0$ and $(-a) + a = 0$.	If $a \neq 0$, there is a unique real number $\frac{1}{a}$, called the reciprocal of a , such that $a \cdot \frac{1}{a} = 1$ and $\frac{1}{a} \cdot a = 1$.	A: $5 + (-5) = 0$ and $(-5) + 5 = 0$ M: $4 \cdot \frac{1}{4} = 1$ and $\frac{1}{4} \cdot 4 = 1$
Distributive	Multiplication distributes over addition.	 $a \cdot (b + c) = a \cdot b + a \cdot c$ $(a + b) \cdot c = a \cdot c + b \cdot c$	$3 \cdot (2 + 5) = 3 \cdot 2 + 3 \cdot 5$ $(2 + 5) \cdot 3 = 2 \cdot 3 + 5 \cdot 3$

Section 1.1 Review. Algebra – fundamental.

Algebra

Properties of Real Numbers

Commutative:	$a + b = b + a; \quad ab = ba$
Associative:	$a + (b + c) = (a + b) + c;$ $a(bc) = (ab)c$
Additive Identity:	$a + 0 = 0 + a = a$
Additive Inverse:	$-a + a = a + (-a) = 0$
Multiplicative Identity:	$a \cdot 1 = 1 \cdot a = a$
Multiplicative Inverse:	$a \cdot \frac{1}{a} = \frac{1}{a} \cdot a = 1, a \neq 0$
Distributive:	$a(b + c) = ab + ac$

Exponents and Radicals

$$a^m \cdot a^n = a^{m+n} \qquad \frac{a^m}{a^n} = a^{m-n}$$

$$(a^m)^n = a^{mn} \qquad (ab)^m = a^m b^m$$

$$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m} \qquad a^{-n} = \frac{1}{a^n}$$

If n is even, $\sqrt[n]{a^n} = |a|$.

If n is odd, $\sqrt[n]{a^n} = a$.

$$\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}, \quad a, b \geq 0$$

$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

$$\sqrt[n]{a^m} = (\sqrt[n]{a})^m = a^{m/n}$$

Factoring Formulas

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^2 + 2ab + b^2 = (a + b)^2$$

$$a^2 - 2ab + b^2 = (a - b)^2$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Interval Notation

$$(a, b) = \{x | a < x < b\}$$

$$[a, b] = \{x | a \leq x \leq b\}$$

$$(a, b] = \{x | a < x \leq b\}$$

$$[a, b) = \{x | a \leq x < b\}$$

$$(-\infty, a) = \{x | x < a\}$$

$$(a, \infty) = \{x | x > a\}$$

$$(-\infty, a] = \{x | x \leq a\}$$

$$[a, \infty) = \{x | x \geq a\}$$

Absolute Value

$$|a| \geq 0$$

For $a > 0$,

$$|X| = a \rightarrow X = -a \quad \text{or} \quad X = a,$$

$$|X| < a \rightarrow -a < X < a,$$

$$|X| > a \rightarrow X < -a \quad \text{or} \quad X > a.$$

Section 1.1

SIDE NOTE

You can remember the order of operations by the acronym **PEMDAS** (Please Excuse My Dear Aunt Sally):

Parentheses
Exponents
Multiplication
Division
Addition
Subtraction.

The Order of Operations

- Step 1** ▶ Whenever a **fraction bar** is encountered, work separately above and below it.
- Step 2** ▶ When **parentheses or other grouping symbols** are present, start inside the innermost pair of grouping symbols and work outward.
- Step 3** ▶ Do operations involving **exponents**.
- Step 4** ▶ Do **multiplications and divisions** in the order in which they occur, **working from left to right**.
- Step 5** ▶ Do **additions and subtractions** in the order in which they occur, **working from left to right**.

Review.

P6 section review.

Square Root Property

For $a \geq 0$, if $x^2 = a$ then $x = \pm\sqrt{a}$.

The Exponent $\frac{1}{n}$

For any real number a and any integer $n > 1$,

$$a^{1/n} = \sqrt[n]{a}.$$

When n is even and $a < 0$, $\sqrt[n]{a}$ and $a^{1/n}$ are not real numbers.

Notice that the denominator n of the rational exponent is the index of the radical.

P6 section review.

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Exponents and Radicals

$$a^m \cdot a^n = a^{m+n}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

$$(a^m)^n = a^{mn}$$

$$(ab)^m = a^m b^m$$

$$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$$

$$a^{-n} = \frac{1}{a^n}$$

If n is even, $\sqrt[n]{a^n} = |a|$.

If n is odd, $\sqrt[n]{a^n} = a$.

$$\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}, \quad a, b \geq 0$$

$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

$$\sqrt[n]{a^m} = (\sqrt[n]{a})^m = a^{m/n}$$

P6 section review.

Square Root Property

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If n is even, $\sqrt[n]{a^n} = |a|$.

If n is odd, $\sqrt[n]{a^n} = a$.

$$\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}, \quad a, b \geq 0$$

$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

$$\sqrt[n]{a^m} = (\sqrt[n]{a})^m = a^{m/n}$$

WARNING

In general:

$$\sqrt[n]{a^n + b^n} \neq a + b$$

$$\sqrt[n]{a + b} \neq \sqrt[n]{a} + \sqrt[n]{b}$$

$\sqrt[n]{a} + \sqrt[n]{b}$ cannot be added if $a \neq b$.

$$\sqrt[n]{a} + \sqrt[n]{a} = 2\sqrt[n]{a}$$

$\sqrt[n]{a} + \sqrt[m]{a}$ cannot be added if $n \neq m$.

P6 section review.

The Exponent $\frac{1}{n}$ and Radicals

If a is any real number and n is any integer greater than 1, then

	n even	n odd
If $a > 0$,	$\sqrt[n]{a} = a^{1/n}$ is positive.	$\sqrt[n]{a} = a^{1/n}$ is positive.
If $a < 0$,	$\sqrt[n]{a} = a^{1/n}$ is not a real number.	$\sqrt[n]{a} = a^{1/n}$ is negative.
If $a = 0$,	$\sqrt[n]{0} = 0^{1/n} = 0$.	$\sqrt[n]{0} = 0^{1/n} = 0$.

Rational Exponents

$$a^{m/n} = (\sqrt[n]{a})^m = \sqrt[n]{a^m},$$

provided that m and n are integers with no common factors, $n > 1$, and $\sqrt[n]{a}$ is a real number.

P6 section review.

Properties of Rational Exponents

If r and s are rational numbers and a and b are real numbers, then

$$\begin{aligned} a^r \cdot a^s &= a^{r+s} & (a^r)^s &= a^{rs} & (ab)^r &= a^r b^r \\ \frac{a^r}{a^s} &= a^{r-s} & \left(\frac{a}{b}\right)^r &= \frac{a^r}{b^r} & a^{-r} &= \frac{1}{a^r} & \left(\frac{a}{b}\right)^{-r} &= \left(\frac{b}{a}\right)^r \end{aligned}$$

provided that all of the expressions used are defined.

Functions; Domain and Range

The temperature at which water boils depends on the elevation above sea level. The interest paid on a cash investment depends on the length of time the investment is held. The area of a circle depends on the radius of the circle. The distance an object travels depends on the elapsed time.

In each case, the value of one variable quantity, say y , depends on the value of another variable quantity, which we often call x . We say that “ y is a function of x ” and write this symbolically as

$$y = f(x) \quad (\text{“}y \text{ equals } f \text{ of } x\text{”}).$$

The symbol f represents the function, the letter x is the **independent variable** representing the input value to f , and y is the **dependent variable** or output value of f at x .

DEFINITION A function f from a set D to a set Y is a rule that assigns a *unique* value $f(x)$ in Y to each x in D .

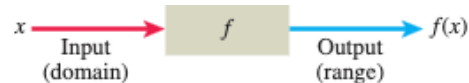


FIGURE 1.1 A diagram showing a function as a kind of machine.

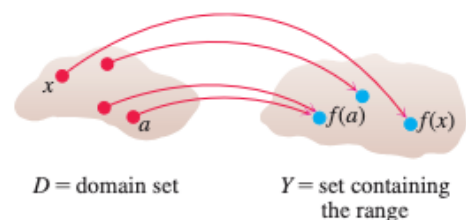


FIGURE 1.2 A function from a set D to a set Y assigns a unique element of Y to each element in D .

EXAMPLE 1 Verify the natural domains and associated ranges of some simple functions. The domains in each case are the values of x for which the formula makes sense.

Function	Domain (x)	Range (y)
$y = x^2$	$(-\infty, \infty)$	$[0, \infty)$
$y = 1/x$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$
$y = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$
$y = \sqrt{4 - x}$	$(-\infty, 4]$	$[0, \infty)$
$y = \sqrt{1 - x^2}$	$[-1, 1]$	$[0, 1]$

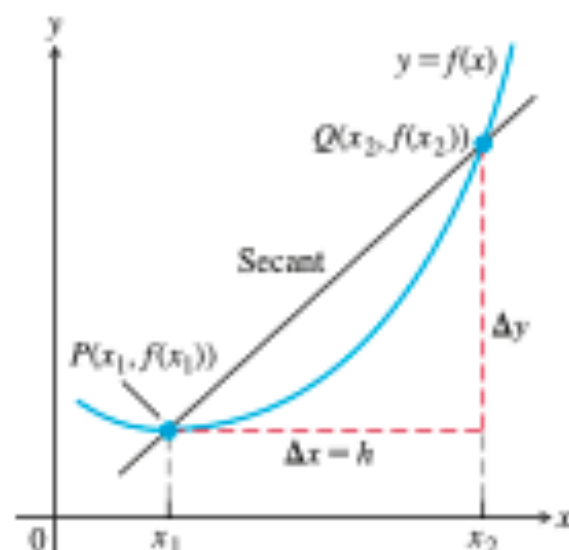


FIGURE 2.1 A secant to the graph $y = f(x)$. Its slope is $\Delta y / \Delta x$, the average rate of change of f over the interval $[x_1, x_2]$.

Average Rates of Change and Secant Lines

Given any function $y = f(x)$, we calculate the average rate of change of y with respect to x over the interval $[x_1, x_2]$ by dividing the change in the value of y , $\Delta y = f(x_2) - f(x_1)$, by the length $\Delta x = x_2 - x_1 = h$ of the interval over which the change occurs. (We use the symbol h for Δx to simplify the notation here and later on.)

DEFINITION The **average rate of change** of $y = f(x)$ with respect to x over the interval $[x_1, x_2]$ is

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{f(x_1 + h) - f(x_1)}{h}, \quad h \neq 0.$$

Geometrically, the rate of change of f over $[x_1, x_2]$ is the slope of the line through the points $P(x_1, f(x_1))$ and $Q(x_2, f(x_2))$ (Figure 2.1). In geometry, a line joining two points of a curve is called a **secant line**. Thus, the average rate of change of f from x_1 to x_2 is identical with the slope of secant line PQ . As the point Q approaches the point P along the curve, the length h of the interval over which the change occurs approaches zero. We will see that this procedure leads to the definition of the slope of a curve at a point.

EXAMPLE 3 Find the slope of the tangent line to the parabola $y = x^2$ at the point $(2, 4)$ by analyzing the slopes of secant lines through $(2, 4)$. Write an equation for the tangent line to the parabola at this point.

Solution We begin with a secant line through $P(2, 4)$ and a nearby point $Q(2 + h, (2 + h)^2)$. We then write an expression for the slope of the secant line PQ and investigate what happens to the slope as Q approaches P along the curve:

$$\begin{aligned}\text{Secant line slope} &= \frac{\Delta y}{\Delta x} = \frac{(2 + h)^2 - 2^2}{h} = \frac{h^2 + 4h + 4 - 4}{h} \\ &= \frac{h^2 + 4h}{h} = h + 4.\end{aligned}$$

If $h > 0$, then Q lies above and to the right of P , as in Figure 2.4. If $h < 0$, then Q lies to the left of P (not shown). In either case, as Q approaches P along the curve, h approaches zero and the secant line slope $h + 4$ approaches 4. We take 4 to be the parabola's slope at P .

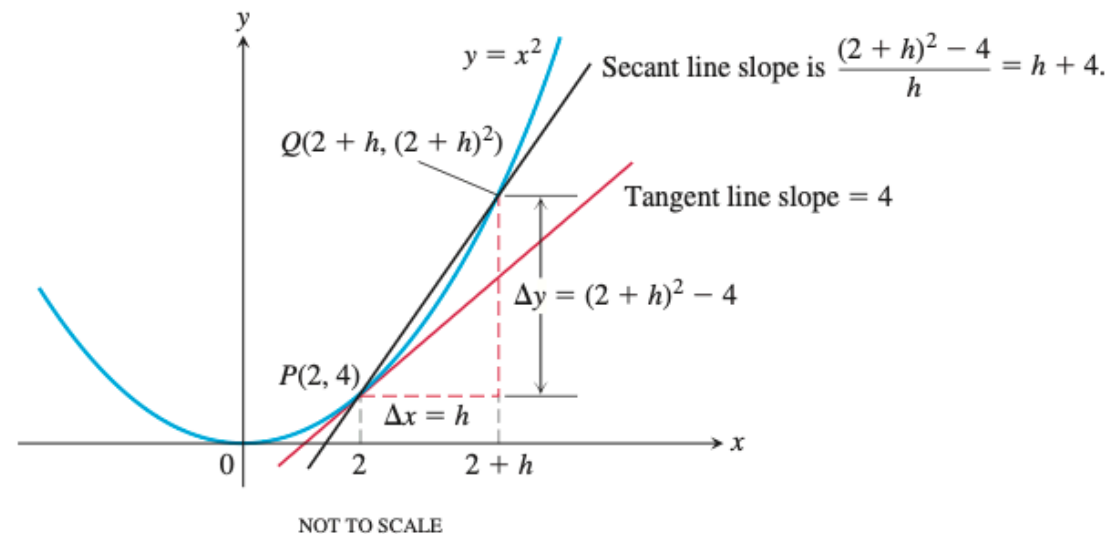


FIGURE 2.4 Finding the slope of the parabola $y = x^2$ at the point $P(2, 4)$ as the limit of secant line slopes (Example 3).

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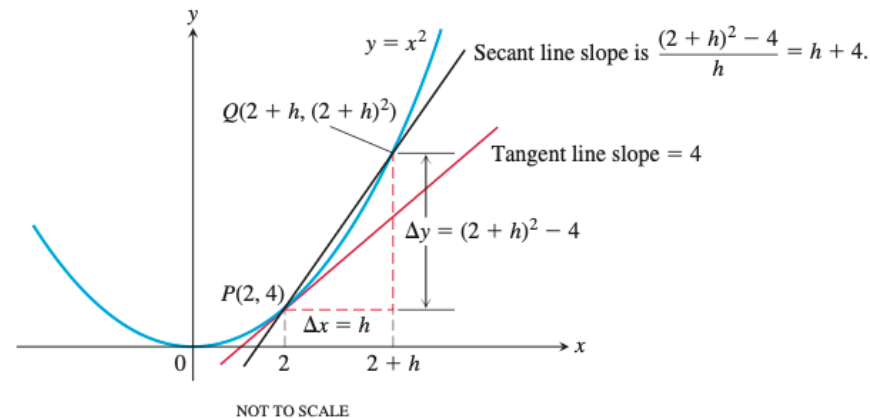


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The tangent line to the parabola at P is the line through P with slope 4:

$$\begin{aligned}y &= 4 + 4(x - 2) && \text{Point-slope equation} \\ y &= 4x - 4.\end{aligned}$$

Slope of a Curve at a Point

In Exercises 7–18, use the method in Example 3 to find **(a)** the slope of the curve at the given point P , and **(b)** an equation of the tangent line at P .

7. $y = x^2 - 5$, $P(2, -1)$

8. $y = 7 - x^2$, $P(2, 3)$

9. $y = x^2 - 2x - 3$, $P(2, -3)$

10. $y = x^2 - 4x$, $P(1, -3)$

11. $y = x^3$, $P(2, 8)$

12. $y = 2 - x^3$, $P(1, 1)$

13. $y = x^3 - 12x$, $P(1, -11)$

14. $y = x^3 - 3x^2 + 4$, $P(2, 0)$

15. $y = \frac{1}{x}$, $P(-2, -1/2)$

An Informal Description of the Limit of a Function

We now give an informal definition of the limit of a function f at an interior point of the domain of f . Suppose that $f(x)$ is defined on an open interval about c , *except possibly at c itself*. If $f(x)$ is arbitrarily close to the number L (as close to L as we like) for all x sufficiently close to c , other than c itself, then we say that f approaches the **limit** L as x approaches c , and write

$$\lim_{x \rightarrow c} f(x) = L,$$

which is read “the limit of $f(x)$ as x approaches c is L .” In Example 1 we would say that $f(x)$ approaches the *limit* 2 as x approaches 1, and write

$$\lim_{x \rightarrow 1} f(x) = 2, \quad \text{or} \quad \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2.$$

Essentially, the definition says that the values of $f(x)$ are close to the number L whenever x is close to c . The value of the function at c itself is not considered.

TABLE 2.2 As x gets closer to 1, $f(x)$ gets closer to 2.

x	$f(x) = \frac{x^2 - 1}{x - 1}$
0.9	1.9
1.1	2.1
0.99	1.99
1.01	2.01
0.999	1.999
1.001	2.001
0.999999	1.999999
1.000001	2.000001

The Limit Laws

A few basic rules allow us to break down complicated functions into simple ones when calculating limits. By using these laws, we can greatly simplify many limit computations.

THEOREM 1 — Limit Laws

If L , M , c , and k are real numbers and

$$\lim_{x \rightarrow c} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = M, \quad \text{then}$$

1. *Sum Rule:* $\lim_{x \rightarrow c} (f(x) + g(x)) = L + M$
2. *Difference Rule:* $\lim_{x \rightarrow c} (f(x) - g(x)) = L - M$
3. *Constant Multiple Rule:* $\lim_{x \rightarrow c} (k \cdot f(x)) = k \cdot L$
4. *Product Rule:* $\lim_{x \rightarrow c} (f(x) \cdot g(x)) = L \cdot M$
5. *Quotient Rule:* $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0$
6. *Power Rule:* $\lim_{x \rightarrow c} [f(x)]^n = L^n, n \text{ a positive integer}$
7. *Root Rule:* $\lim_{x \rightarrow c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}, n \text{ a positive integer}$

(If n is even, we assume that $f(x) \geq 0$ for x in an interval containing c .)

EXAMPLE 5 Use the observations $\lim_{x \rightarrow c} k = k$ and $\lim_{x \rightarrow c} x = c$ (Example 3) and the limit laws in Theorem 1 to find the following limits.

(a) $\lim_{x \rightarrow c} (x^3 + 4x^2 - 3)$

(b) $\lim_{x \rightarrow c} \frac{x^4 + x^2 - 1}{x^2 + 5}$

(c) $\lim_{x \rightarrow -2} \sqrt{4x^2 - 3}$

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Solution

$$\begin{aligned} \text{(a)} \quad \lim_{x \rightarrow c} (x^3 + 4x^2 - 3) &= \lim_{x \rightarrow c} x^3 + \lim_{x \rightarrow c} 4x^2 - \lim_{x \rightarrow c} 3 \\ &= c^3 + 4c^2 - 3 \end{aligned}$$

Sum and Difference Rules

Power and Multiple Rules

$$\text{(b)} \quad \lim_{x \rightarrow c} \frac{x^4 + x^2 - 1}{x^2 + 5} = \frac{\lim_{x \rightarrow c} (x^4 + x^2 - 1)}{\lim_{x \rightarrow c} (x^2 + 5)}$$

Quotient Rule

$$= \frac{\lim_{x \rightarrow c} x^4 + \lim_{x \rightarrow c} x^2 - \lim_{x \rightarrow c} 1}{\lim_{x \rightarrow c} x^2 + \lim_{x \rightarrow c} 5}$$

Sum and Difference Rules

$$= \frac{c^4 + c^2 - 1}{c^2 + 5}$$

Power or Product Rule

$$\text{(c)} \quad \lim_{x \rightarrow -2} \sqrt{4x^2 - 3} = \sqrt{\lim_{x \rightarrow -2} (4x^2 - 3)}$$

Root Rule with $n = 2$

$$= \sqrt{\lim_{x \rightarrow -2} 4x^2 - \lim_{x \rightarrow -2} 3}$$

Difference Rule

$$= \sqrt{4(-2)^2 - 3}$$

Product and Multiple Rules and limit of a constant function

$$= \sqrt{16 - 3}$$

$$= \sqrt{13}$$



Calculating Limits

Find the limits in Exercises 11–22.

11. $\lim_{x \rightarrow -3} (x^2 - 13)$

12. $\lim_{x \rightarrow 2} (-x^2 + 5x - 2)$

13. $\lim_{t \rightarrow 6} 8(t - 5)(t - 7)$

14. $\lim_{x \rightarrow -2} (x^3 - 2x^2 + 4x + 8)$

15. $\lim_{x \rightarrow 2} \frac{2x + 5}{11 - x^3}$

16. $\lim_{s \rightarrow 2/3} (8 - 3s)(2s - 1)$

17. $\lim_{x \rightarrow -1/2} 4x(3x + 4)^2$

18. $\lim_{y \rightarrow 2} \frac{y + 2}{y^2 + 5y + 6}$

19. $\lim_{y \rightarrow -3} (5 - y)^{4/3}$

20. $\lim_{z \rightarrow 4} \sqrt{z^2 - 10}$

- Questions?

Study hard, best of luck!